

## Compressing LLMs using Distillation

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### Project Description:

Our project introduces three pivotal papers and a hands-on demo, each addressing distinct approaches to compressing LLMs for enhanced efficiency, performance:

1. **Distilling Step by Step (Paper 1):** This method leverages rationales as supervision during training, where teacher-generated step-by-step reasoning guides student models. By treating rationales as inputs or supervision, smaller models achieve superior performance while requiring significantly less data. To validate this, we conducted a **demo** using the CQA dataset labeled with PaLM. The accuracy improvements observed were as follows:
  - **CQA Test Set on T5 Small (No Distillation):** 0% Accuracy (random completions).
  - **After Standard Distillation:** 14.08% Accuracy after 5 epochs.
  - **After Step-by-Step Distillation:** 25.47% Accuracy after 5 epochs.
2. **MiniLLM (Paper 2):** Focuses on optimizing Reverse KL Divergence for knowledge distillation, addressing Forward KL's limitations in open-text generation. The study also introduces techniques like length normalization, teacher-mixed sampling, and single-step decomposition to improve student models' alignment with teacher distributions.
3. **Minitron (Paper 3):** Combines structured pruning (width and depth) with teacher correction to create smaller, high-performing models. These methods enable significant reductions in training tokens (up to 150×) and runtime speedups (up to 2.7×) while preserving or enhancing accuracy.

Together, these approaches and our demo showcase how LLMs can be effectively scaled by reducing computational demands and enabling deployment on resource-constrained systems without sacrificing performance or accuracy.